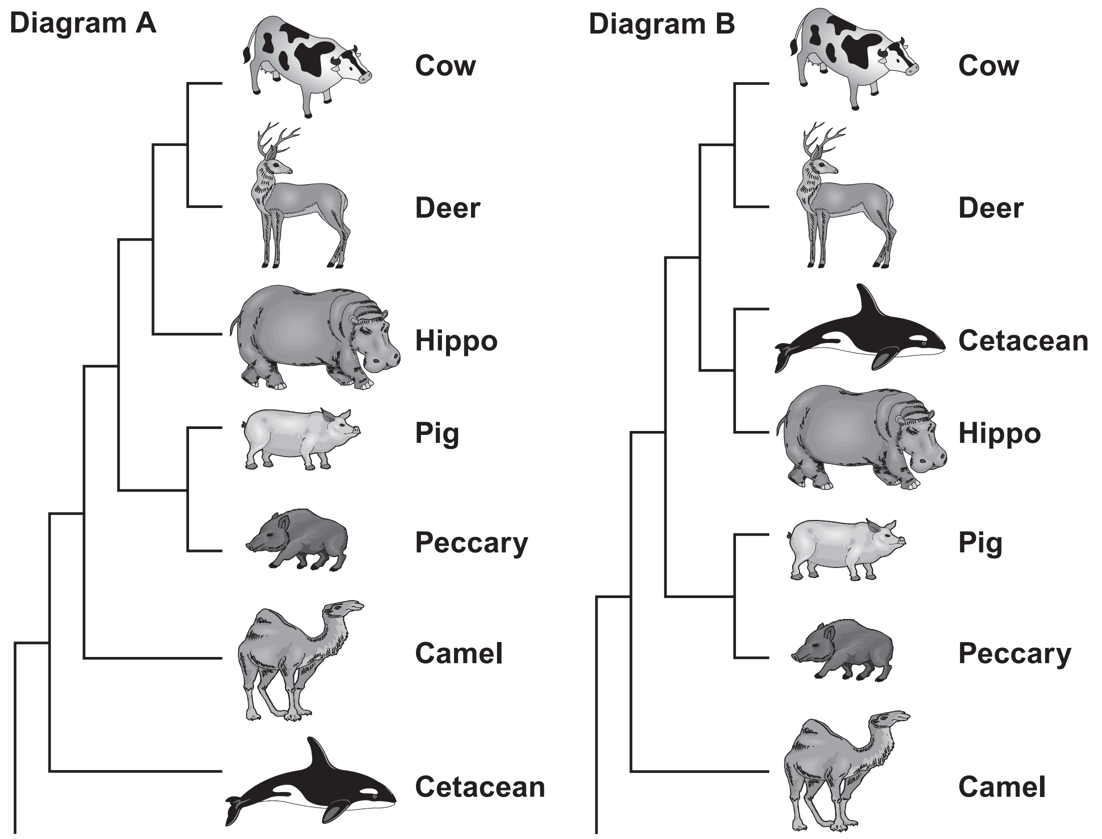
3. Whales and dolphins belong to a single group of carnivorous, marine mammals called the cetaceans (order Cetacea). Cetaceans are comprised of three sub-orders: Odontoceti (toothed whales including sperm whales and dolphins), Mysticeti (baleen whales), and Archaeoceti (the extinct ancestors of modern whales).

There have been a number of theories regarding the closest living relative to the cetaceans.

The diagrams below illustrate two of these theories. With the exception of the cetacean, all the mammals shown belong to the order Artiodactyla.



(a) State the term used to describe diagrams such as those shown above. [1]

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(b) The values given in the following table show the number of differences in the nucleotide sequence of the gene coding for the synthesis of the milk protein casein in different mammals.

Sperm whale	3							
Dolphin	3	2						
Hippo	4	3	3					
Cow	9	8	8	8				
Camel	12	11	11	12	14			
Deer	11	10	10	10	4	16		
Pig	11	10	10	11	13	14	13	
Peccary	14	12	13	14	16	16	18	7
	Baleen whale	Sperm whale	Dolphin	Hippo	Cow	Camel	Deer	Pig

(i) Use the information in the table to explain whether **Diagram A** or **Diagram B** represents the currently accepted theory regarding the closest living relative to the cetaceans. [3]

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(ii) Modern taxonomic classification combines Cetacea and Artiodactyla into a single order, the Cetiartiodactyla. Explain how this illustrates the "tentative nature" of biological classification. [2]

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(c) Both the common bottlenose dolphin *(Tursiops truncates)* and the killer whale *(Orcinis orca)* belong to a smaller taxonomic group of the sub-order Odontoceti called the Delphinidae. Name the group in the taxonomic hierarchy to which the Delphinidae belong. [1]

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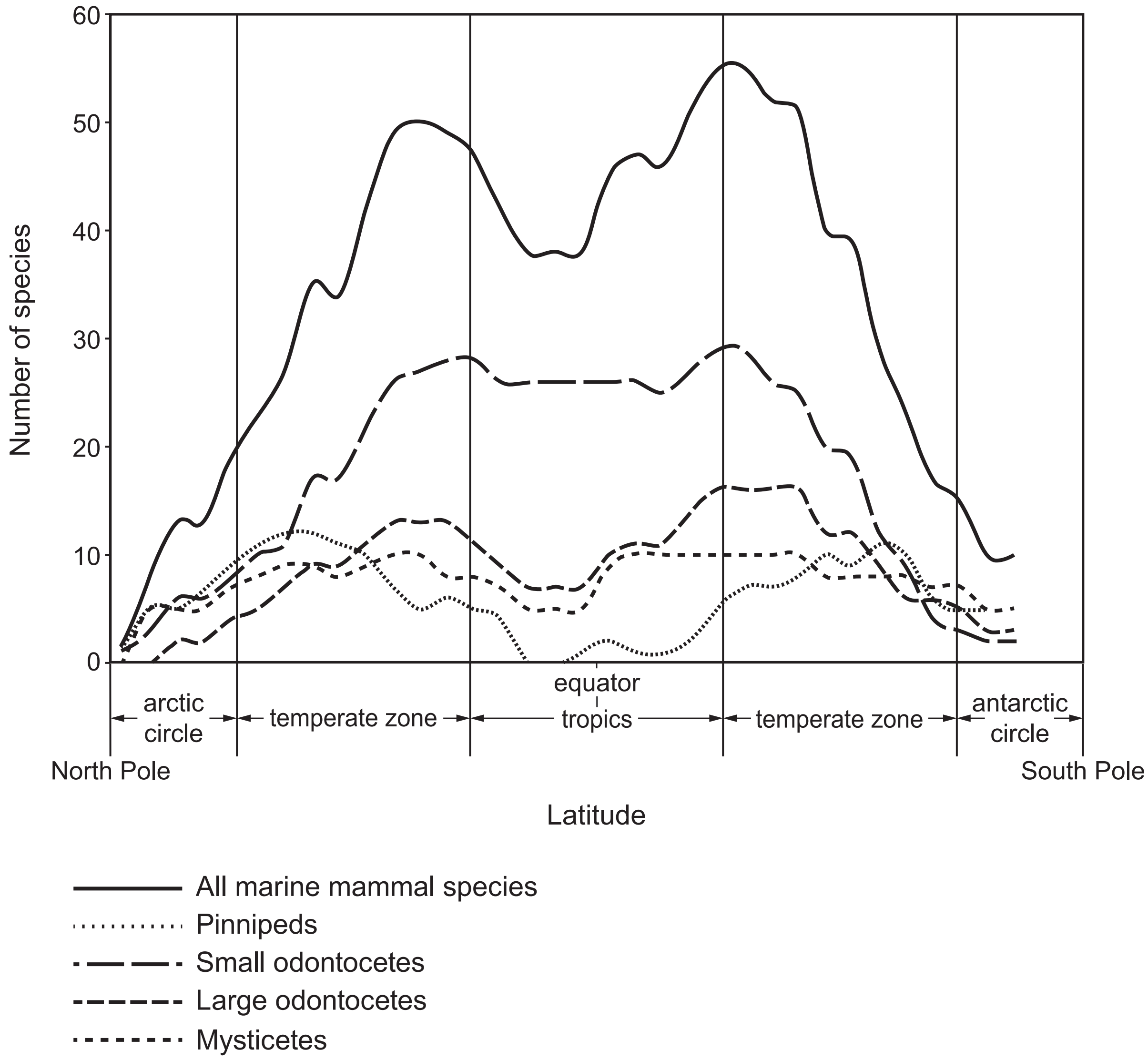
(d) In 2011, an international group of researchers used sightings from three oceanic surveys to predict patterns in the global distribution of marine mammals. The table lists the mammalian groups included in the survey.

Mammalian group	Examples
Pinnipeds	seals and sea lions
Small odontocetes	dolphins
Large odontocetes	sperm whales and killer whales
Mysticetes	baleen whales

The following graph shows the predicted number of species by latitude.



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Examiner only

(i) Describe the effect of latitude on the number of species of small odontocetes from the **antarctic circle** to the **tropics**. [2]

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(ii) State the environmental factor that is **most** likely to explain the distribution of all marine mammal species. [1]

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(iii) Why is the curve for **all marine mammal species** described as showing a bimodal distribution? [1]

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